Multimodal Misinformation Detector Agent

Data Science and AI Lab Project

Milestone 3 Report

Group 9

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Contents

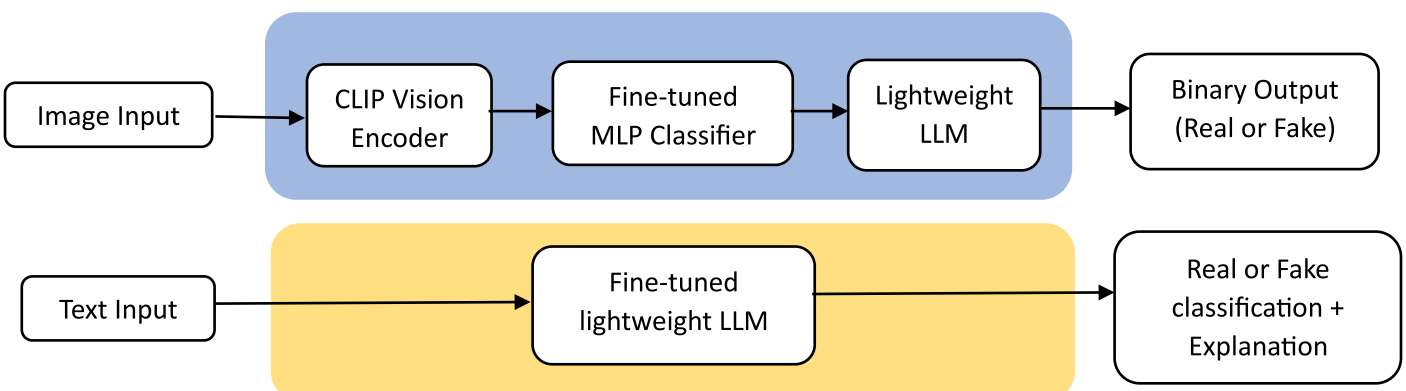
1. Objective

The objective of this milestone is to present multimodal misinformation detection architecture, detailing its design, components, and operational workflows. Two independent architectures are proposed hereunder which include separate pipelines for image and text inputs. Clear justifications for design choices are also highlighted.

2. Model Architecture - 1

2.1 Overview

The proposed system consists of two independent pipelines for image and text inputs. The architecture leverages pre-trained models with minimal fine-tuning to ensure lightweight deployment, maintain interpretability, and provide confidence estimates along with human-readable explanations.

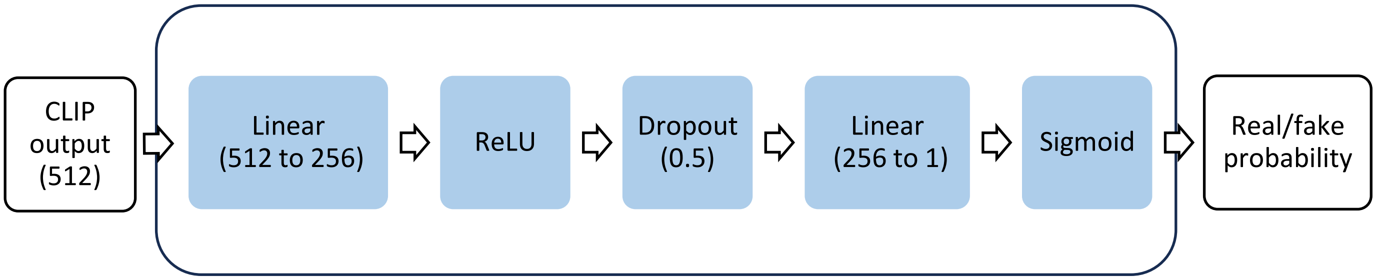


2.2 Image pipeline

The image pipeline processes input images to determine authenticity as real or fake and provides a short explanatory text. It has below components:

- **CLIP Vision Encoder:** openai/clip-vit-base-patch32 model to extract image embeddings (512-dimensional vectors) that encode rich visual features, enabling effective real/fake image classification.

- **Fine-tuned MLP Classifier :** This maps CLIP embeddings to real/fake probability. It follows the below architecture:



- **Lightweight LLM:** microsoft/phi-3-mini-4k-instruct or similar model will be used to take model prediction and confidence score as input and give one-to-two sentence explanation of why an image may appear fake or real.

- **UI Layer:** Built with Gradio, accepts image input, displays authenticity label, and confidence score.

2.3 Text pipeline

The text pipeline evaluates the credibility of textual claims independently from images. It has below components:

- **Lightweight LLM:** microsoft/phi-3-mini-4k-instruct predicts the credibility label (True, False, Partly True) and simultaneously generates a concise, human-readable explanation. Fine-tuning on FEVER datasets (claims with rationales) can improve label accuracy and reasoning quality, but zero-shot or few-shot prompting is feasible.

- **UI Layer:** Integrated with the image module in Gradio, displaying the label and human readable explanation.

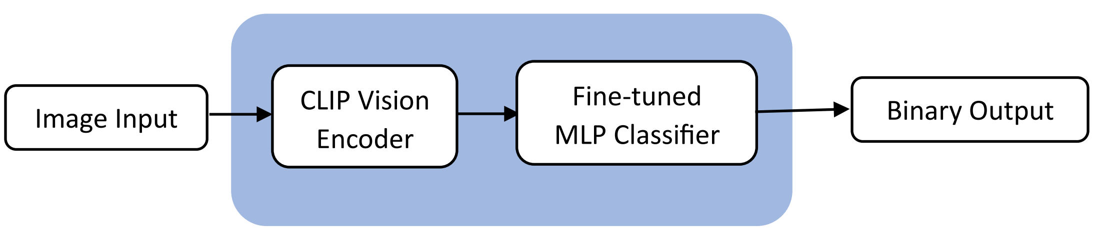
2.4 Justification

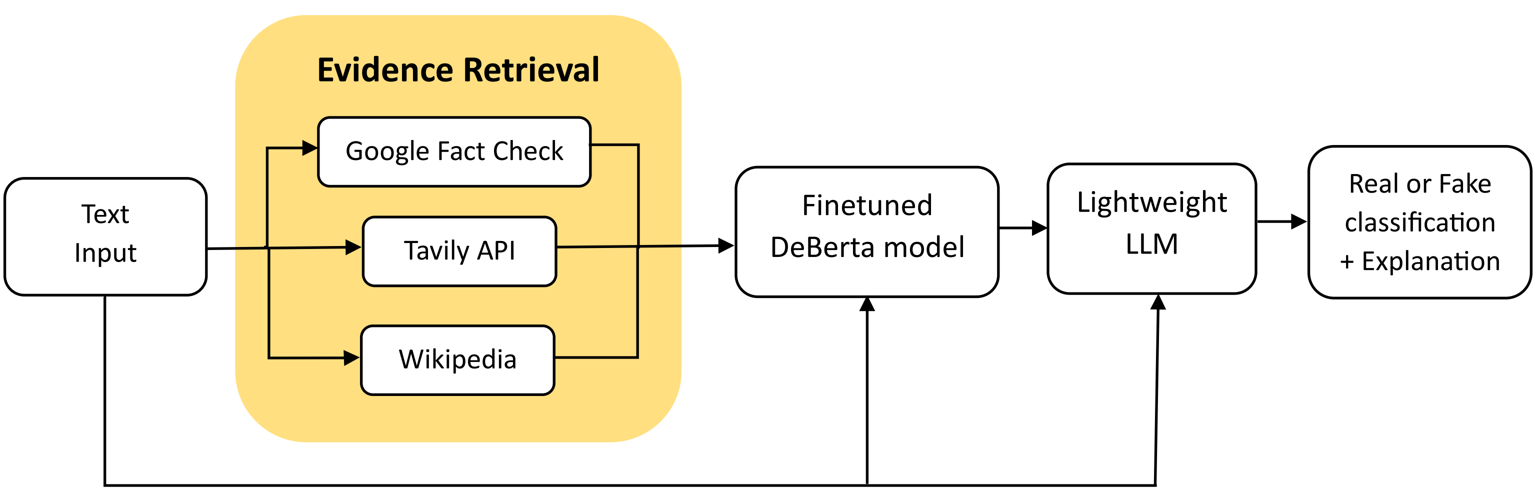
- Images and text are processed independently because the input modalities are not semantically aligned, simplifying the model design.
- With powerful pre-training, CLIP extracts meaningful visual features. They work well for fine-tuning on small datasets. They can run efficiently even without GPU.
- Small parameter count in MLP classifier allows fast training.
- Phi-3-mini allows direct instruction-based explanations.

3. Model Architecture - 2

3.1 Overview

Each modality (image and text) remains independent, but predictions are augmented or validated via external Web APIs. This enables real-time access to up-to-date knowledge, metadata, and image verification services, improving reliability and explainability without combining embeddings.





3.2 Image pipeline

- **Image Processing**: openai/clip-vit-base-patch32 extracts image embeddings (512-dimensional vectors) that encode rich visual features for real/fake classification.
- **Fine-tuned MLP Classifier :** This maps CLIP embeddings to real/fake probability.
- **UI Layer:** Gradio interface displays the authenticity label, confidence score, explanation, and optional Web API results.

3.3 Text Pipeline

- **Web API Fact-Checking:** Fact-checking API from Tavily, Google Fact Check and Wikipedia will provide external verification and supporting evidence.
- **Text Processing:** Fine-tuned Deberta model takes the textual claim and predicts the credibility label (True, or False) based on supporting evidences
- **Explanation Generation:** LLM integrates model predictions with input claim to generate concise, human-readable explanations referencing external sources.
- **UI Layer:** Gradio interface shows the credibility label, confidence score, explanation, and optionally source links.

3.4 Justification

- Images and text are processed independently because the input modalities are not semantically aligned, simplifying the model design.
- Web API integration provides real-time verification and external evidence, improving reliability without increasing model complexity.
- Local models (CLIP for images, phi-3-mini for text) extract meaningful embeddings that work well with small datasets and lightweight fine-tuning.
- Small parameter count in MLP classifier allows fast training.

- Phi-3-mini generates human-readable explanations that combine model predictions with Web API results, enhancing transparency and interpretability.

4. Comparison

Below is a concise comparison of the models discussed above.

Feature	Model -1	Model -2
Image Pipeline	CLIP Vision Encoder + MLP classifier	CLIP Vision Encoder + MLP classifier
Text Pipeline	LLM (phi-3-mini)	Fact-checking APIs + Finetuned Deberta + mistralai
Explanations	Generated by LLM based on local model outputs	Generated by LLM using input claim + model prediction
Reliability	Depends solely on local model performance	Enhanced with real-time external verification
Inference Cost	Low, runs efficiently on CPU/GPU	Slightly higher due to API calls but still lightweight
Use Case Suitability	Rapid deployment, offline scenarios	Scenarios requiring up-to-date verification, external evidence

5. Conclusion

In this milestone, we have presented and analyzed two independent multimodal architectures for misinformation detection. The comparison highlights trade-offs between deployment simplicity and enhanced verification capabilities, allowing us to evaluate the approaches and decide on the most suitable architecture in upcoming development stages.